

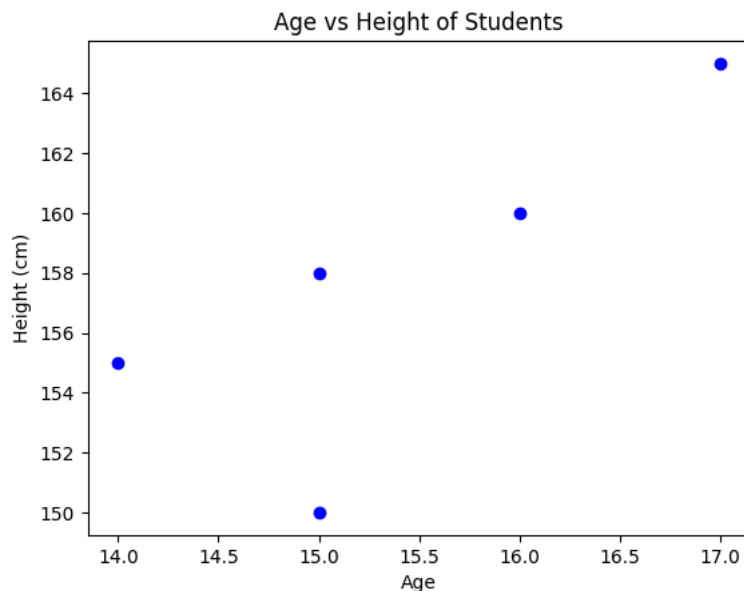
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- ROLL NO : S008
- CLASS : SYIT
- SUB : Python for DS
- DATE OF ASSIGNMENT : 02-09-26
- DATE OF SUBMISSION : 02-09-26

```
import matplotlib.pyplot as plt
import numpy as np
```

a. Write a Python program to create a scatter plot of Age and Height of 5 students.

```
ages = [15, 16, 14, 17, 15]
heights = [150, 160, 155, 165, 158]

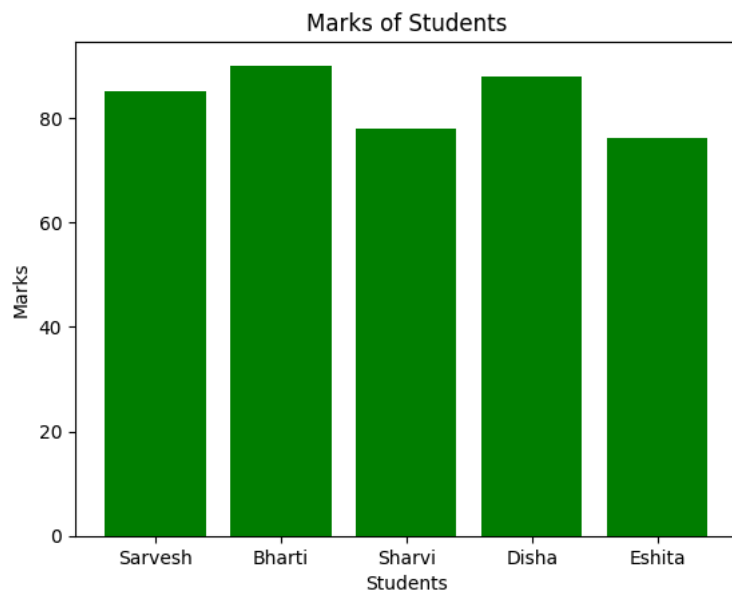
plt.scatter(ages, heights, color='blue')
plt.title("Age vs Height of Students")
plt.xlabel("Age")
plt.ylabel("Height (cm)")
plt.show()
```



b. Write a Python program to create a bar chart showing marks of 5 students.

```
students = ["Sarvesh", "Bharti", "Sharvi", "Disha", "Eshita"]
marks = [85, 90, 78, 88, 76]

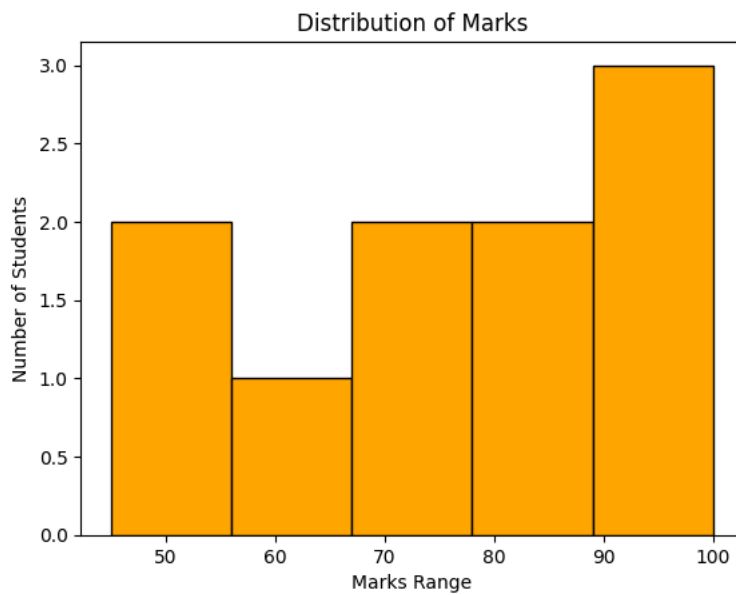
plt.bar(students, marks, color='green')
plt.title("Marks of Students")
plt.xlabel("Students")
plt.ylabel("Marks")
plt.show()
```



c. Write a Python program to create a histogram showing the distribution of marks of 10 students.

```
marks = [45, 55, 65, 70, 75, 80, 85, 90, 95, 100]

plt.hist(marks, bins=5, color='orange', edgecolor='black')
plt.title("Distribution of Marks")
plt.xlabel("Marks Range")
plt.ylabel("Number of Students")
plt.show()
```

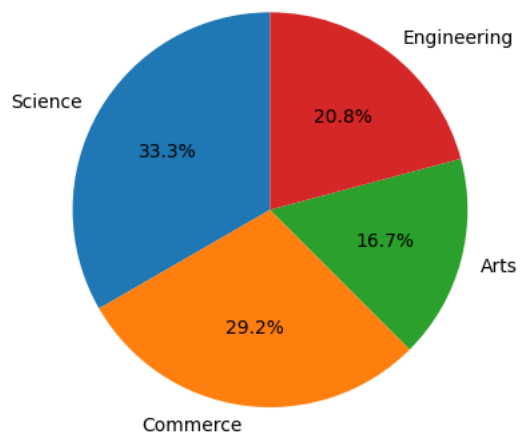


d. Write a Python program to create a pie chart showing the number of students in different courses.

```
courses = ["Science", "Commerce", "Arts", "Engineering"]
students = [40, 35, 20, 25]

plt.pie(students, labels=courses, autopct='%1.1f%%', startangle=90)
plt.title("Students in Different Courses")
plt.show()
```

Students in Different Courses



e. Write a Python program to create a bar chart showing sales of 5 different products.

```
products = ["Laptop", "Smartphone", "Headphones", "Smartwatch", "Tablet"]
sales = [500, 700, 400, 650, 800]

plt.bar(products, sales, color='pink')
plt.title("Sales of Products")
plt.xlabel("Products")
plt.ylabel("Sales")
plt.show()
```

